

CODING CHALLENGE 03*Testing Report***I. C01: Loan Decision Engine****Test Case 01: An approved application**

Input values used: Monthly Income 50000, Monthly Debt 10000, Credit Score 760, Employment Duration 24, Requested Loan 300000, Existing Customer yes

Expected decision or mathematical result: DTI = 20.00% (Low), credit is Excellent, and the maximum loan is PHP 600,000, so the request is within the limit. APPROVED, because credit is Excellent or Good, $DTI \leq 40$, and employment ≥ 12 months.

Actual program output:

Python

```
=====
                        LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 50000
Monthly Debt (PHP): 10000
Credit Score: 760
Employment Duration (months): 24
Requested Loan (PHP): 300000
Existing Customer (yes/no): yes

=====
                        LOAN DECISION ASSESSMENT
=====
Monthly Income:           PHP 50,000.00
Monthly Debt:             PHP 10,000.00
Requested Loan:           PHP 300,000.00
Employment Duration:      24 months
Credit Score:             760
Existing Customer:        Yes
-----
ASSESSMENT
-----
Credit Classification:    Excellent
Debt-to-Income Ratio:     20.00%
```

DTI Classification: Low
Maximum Recommended Loan: PHP 600,000.00
Requested Amount Status: Within Recommended Limit

FINAL DECISION

APPROVED

Reason: Credit, DTI, and employment duration satisfy the approval rules.

```
Run CC03_Challenge01_LoanDecisionEngine
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge01_LoanDecisionEngine.py

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 50000
Monthly Debt (PHP): 10000
Credit Score: 760
Employment Duration (months): 24
Requested Loan (PHP): 300000
Existing Customer (yes/no): yes

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income: PHP 50,000.00
Monthly Debt: PHP 10,000.00
Requested Loan: PHP 300,000.00
Employment Duration: 24 months
Credit Score: 760
Existing Customer: Yes

=====
ASSESSMENT
=====
Credit Classification: Excellent
Debt-to-Income Ratio: 20.00%
DTI Classification: Low
Maximum Recommended Loan: PHP 600,000.00

=====
FINAL DECISION
=====
APPROVED
Reason: Credit, DTI, and employment duration satisfy the approval rules.
=====

Process finished with exit code 0
```

PASSED

Test Case 02: An application for review

Input values used: Monthly Income 40000, Monthly Debt 14000, Credit Score 680, Employment Duration 18, Requested Loan 200000, Existing Customer no

Expected decision or mathematical result: DTI = 35.00% (Moderate), credit is Fair, and the request is within the PHP 480,000 limit. FOR REVIEW, because Fair credit with $DTI \leq 40$ and employment ≥ 12 months requires additional review.

Actual program output:

```
Python
=====
                        LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 40000
Monthly Debt (PHP): 14000
Credit Score: 680
Employment Duration (months): 18
Requested Loan (PHP): 200000
Existing Customer (yes/no): no

=====
                        LOAN DECISION ASSESSMENT
=====
Monthly Income:           PHP 40,000.00
Monthly Debt:             PHP 14,000.00
Requested Loan:           PHP 200,000.00
Employment Duration:      18 months
Credit Score:            680
Existing Customer:        No
-----
ASSESSMENT
-----
Credit Classification:    Fair
Debt-to-Income Ratio:    35.00%
DTI Classification:       Moderate
Maximum Recommended Loan: PHP 480,000.00
Requested Amount Status:  Within Recommended Limit
-----
FINAL DECISION
-----
FOR REVIEW
Reason: Fair credit classification requires additional review.
=====
```

```
Run CC03_Challenge01_LoanDecisionEngine
/Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/venv/bin/python /Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge01_LoanDecisionEngine.py

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 40000
Monthly Debt (PHP): 14000
Credit Score: 680
Employment Duration (months): 18
Requested Loan (PHP): 200000
Existing Customer (yes/no): no

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income:      PHP 40,000.00
Monthly Debt:        PHP 14,000.00
Requested Loan:       PHP 200,000.00
Employment Duration: 18 months
Credit Score:        680
Existing Customer:    No

=====
ASSESSMENT
=====
Credit Classification: Fair
Debt-to-Income Ratio: 35.00%
DTI Classification: Moderate
Maximum Recommended Loan: PHP 480,000.00

=====
Existing Customer (yes/no): no

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income:      PHP 40,000.00
Monthly Debt:        PHP 14,000.00
Requested Loan:       PHP 200,000.00
Employment Duration: 18 months
Credit Score:        680
Existing Customer:    No

=====
ASSESSMENT
=====
Credit Classification: Fair
Debt-to-Income Ratio: 35.00%
DTI Classification: Moderate
Maximum Recommended Loan: PHP 480,000.00
Requested Amount Status: Within Recommended Limit

=====
FINAL DECISION
=====
FOR REVIEW
Reason: Fair credit classification requires additional review.
=====
Process finished with exit code 0
```

PASSED

Test Case 03: A declined application

Input values used: Monthly Income 30000, Monthly Debt 18000, Credit Score 720, Employment Duration 24, Requested Loan 100000, Existing Customer yes

Expected decision or mathematical result: DTI = 60.00% (Very High) and credit is Good. DECLINED, reason: "Debt-to-income ratio is above 50%."

Actual program output:

```
Python
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 30000
Monthly Debt (PHP): 18000
Credit Score: 720
Employment Duration (months): 24
Requested Loan (PHP): 100000
```

Existing Customer (yes/no): yes

=====

LOAN DECISION ASSESSMENT

=====

Monthly Income: PHP 30,000.00
Monthly Debt: PHP 18,000.00
Requested Loan: PHP 100,000.00
Employment Duration: 24 months
Credit Score: 720
Existing Customer: Yes

ASSESSMENT

Credit Classification: Good
Debt-to-Income Ratio: 60.00%
DTI Classification: Very High
Maximum Recommended Loan: PHP 360,000.00
Requested Amount Status: Within Recommended Limit

FINAL DECISION

DECLINED

Reason: Debt-to-income ratio is above 50%.

=====

```
Run CC03_Challenge01_LoanDecisionEngine x
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge01_LoanDecis

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 30000
Monthly Debt (PHP): 18000
Credit Score: 720
Employment Duration (months): 24
Requested Loan (PHP): 100000
Existing Customer (yes/no): yes

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income: PHP 30,000.00
Monthly Debt: PHP 18,000.00
Requested Loan: PHP 100,000.00
Employment Duration: 24 months
Credit Score: 720
Existing Customer: Yes

-----
ASSESSMENT
-----
Credit Classification: Good
Debt-to-Income Ratio: 60.00%
DTI Classification: Very High
Maximum Recommended Loan: PHP 360,000.00
```

```
Run CC03_Challenge01_LoanDecisionEngine
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income:      PHP 30,000.00
Monthly Debt:        PHP 10,000.00
Requested Loan:      PHP 100,000.00
Employment Duration: 24 months
Credit Score:        720
Existing Customer:    Yes
=====
ASSESSMENT
=====
Credit Classification: Good
Debt-to-Income Ratio:  60.00%
DTI Classification:    Very High
Maximum Recommended Loan: PHP 360,000.00
Requested Amount Status: Within Recommended Limit
=====
FINAL DECISION
=====
DECLINED
Reason: Debt-to-income ratio is above 50%.
=====
Process finished with exit code 0
```

PASSED

Test Case 04: An invalid input case

Input values used: Monthly Income 0, Monthly Debt 5000, Credit Score 700, Employment Duration 12, Requested Loan 100000, Existing Customer no

Expected decision or mathematical result: The program prints INVALID INPUT with the reason "Monthly income must be greater than 0." and "No approval decision can be made." No DTI or decision is computed.

Actual program output:

```
Python
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 0
Monthly Debt (PHP): 5000
Credit Score: 700
Employment Duration (months): 12
Requested Loan (PHP): 100000
Existing Customer (yes/no): no

-----
INVALID INPUT
-----
Reason: Monthly income must be greater than 0.
No approval decision can be made.
=====
```

```
Run CC03_Challenge01_LoanDecisionEngine x
/Users/francisobille/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge01_LoanDecis3
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 0
Monthly Debt (PHP): 5000
Credit Score: 700
Employment Duration (months): 12
Requested Loan (PHP): 100000
Existing Customer (yes/no): no

-----
INVALID INPUT
-----
Reason: Monthly income must be greater than 0.
No approval decision can be made.
=====
```

PASSED

Test Case 05: At least one boundary-value case

Input values used: Monthly Income 50000, Monthly Debt 20000, Credit Score 750, Employment Duration 12, Requested Loan 600000, Existing Customer yes

Expected decision or mathematical result:

This case sits on four boundaries at once:

- Credit score 750 is exactly Excellent.
- DTI is exactly 40.00%, classified Moderate and still ≤ 40 .
- Employment is exactly 12 months.
- The request equals the maximum loan ($12 \times \text{income}$), which is "Within Recommended Limit" because the check uses $\leq$.
- Result: APPROVED.

Actual program output:

```
Python
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 50000
Monthly Debt (PHP): 20000
Credit Score: 750
Employment Duration (months): 12
Requested Loan (PHP): 600000
Existing Customer (yes/no): yes

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income:          PHP 50,000.00
Monthly Debt:            PHP 20,000.00
```

Requested Loan: PHP 600,000.00
Employment Duration: 12 months
Credit Score: 750
Existing Customer: Yes

ASSESSMENT

Credit Classification: Excellent
Debt-to-Income Ratio: 40.00%
DTI Classification: Moderate
Maximum Recommended Loan: PHP 600,000.00
Requested Amount Status: Within Recommended Limit

FINAL DECISION

APPROVED

Reason: Credit, DTI, and employment duration satisfy the approval rules.

```
Run CC03_Challenge01_LoanDecisionEngine
/Users/francisobille/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge01_LoanDecisionEngine.py

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income (PHP): 50000
Monthly Debt (PHP): 20000
Credit Score: 750
Employment Duration (months): 12
Requested Loan (PHP): 600000
Existing Customer (yes/no): yes

=====
LOAN DECISION ASSESSMENT
=====
Monthly Income: PHP 50,000.00
Monthly Debt: PHP 20,000.00
Requested Loan: PHP 600,000.00
Employment Duration: 12 months
Credit Score: 750
Existing Customer: Yes

=====
ASSESSMENT
=====
Credit Classification: Excellent
Debt-to-Income Ratio: 40.00%
DTI Classification: Moderate
Maximum Recommended Loan: PHP 600,000.00
```

```
Run CC03_Challenge01_LoanDecisionEngine
=====
EXISTING CUSTOMER: YES/NO: yes
=====
LOAN DECISION ASSESSMENT
=====
Monthly Income:      PHP 50,000.00
Monthly Debt:        PHP 20,000.00
Requested Loan:      PHP 600,000.00
Employment Duration: 12 months
Credit Score:       750
Existing Customer:   Yes
=====
ASSESSMENT
=====
Credit Classification: Excellent
Debt-to-Income Ratio:  40.00%
DTI Classification:    Moderate
Maximum Recommended Loan: PHP 600,000.00
Requested Amount Status: Within Recommended Limit
=====
FINAL DECISION
=====
APPROVED
Reason: Credit, DTI, and employment duration satisfy the approval rules.
=====
```

PASSED

II. C02: Quadratic Analyzer

Test Case 01: Positive discriminant

Input values used: $a = 1$, $b = -3$, $c = 2$

Expected decision or mathematical result: Discriminant = 1 (POSITIVE), so two distinct real roots: $x_1 = 2.00$ and $x_2 = 1.00$. Vertex is (1.50, -0.25). The parabola crosses the x-axis twice.

Actual program output:

```
Python
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 1
b: -3
c: 2

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
1.00x^2 + -3.00x + 2.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.50, -0.25)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 1.00
```

Classification: POSITIVE

ROOTS CLASSIFICATION

Number of Solutions: TWO DISTINCT REAL ROOTS

Geometric Interpretation: The parabola crosses the x-axis twice.

SOLUTIONS

Root 1 (x_1) = 2.00

Root 2 (x_2) = 1.00

x-intercepts: (2.00, 0) and (1.00, 0)

```
Run CC03_Challenge02_QuadraticAnalyzer x
/Users/francisobille/Documents/Coding-Challenge03-APPOAET/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET/CC03_Challenge02_QuadraticAnalyzer.py
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 1
b: -3
c: 2

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
1.00x^2 + -3.00x + 2.00 = 0

PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.50, -0.25)

DISCRIMINANT ANALYSIS
-----
Discriminant Value: 1.00
Classification: POSITIVE

ROOTS CLASSIFICATION
-----
Number of Solutions: TWO DISTINCT REAL ROOTS
Geometric Interpretation: The parabola crosses the x-axis twice.

=====
SOLUTIONS
-----
Root 1 (x1) = 2.00
Root 2 (x2) = 1.00
x-intercepts: (2.00, 0) and (1.00, 0)
=====
```

PASSED

Test Case 02: Zero discriminant

Input values used: $a = 1$, $b = -2$, $c = 1$

Expected decision or mathematical result: Discriminant = 0 (ZERO), so one repeated real root: $x = 1.00$. Vertex is $(1.00, 0.00)$, resting exactly on the x-axis.

Actual program output:

Python

Enter coefficients for the quadratic equation ($ax^2 + bx + c = 0$):

a: 1

b: -2

c: 1

=====

QUADRATIC EQUATION AND PARABOLA ANALYZER

=====

EQUATION

$1.00x^2 + -2.00x + 1.00 = 0$

PARABOLA PROPERTIES

Opening Direction: UPWARD

Vertex Type: MINIMUM

Vertex Coordinates: $(1.00, 0.00)$

DISCRIMINANT ANALYSIS

Discriminant Value: 0.00

Classification: ZERO

ROOTS CLASSIFICATION

Number of Solutions: ONE REAL ROOT (REPEATED)

Geometric Interpretation: The parabola's vertex rests exactly on the x-axis.

SOLUTION

Root (x) = 1.00

x-intercept: $(1.00, 0)$

=====

```
Run CC03_Challenge02_QuadraticAnalyzer x
/Users/francisobille/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge02_QuadraticAn
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 1
b: -2
c: 1
=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
1.00x^2 + -2.00x + 1.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.00, 0.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 0.00
Classification: ZERO
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: ONE REAL ROOT (REPEATED)
Geometric Interpretation: The parabola's vertex rests exactly on the x-axis.

=====
EQUATION
1.00x^2 + -2.00x + 1.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.00, 0.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 0.00
Classification: ZERO
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: ONE REAL ROOT (REPEATED)
Geometric Interpretation: The parabola's vertex rests exactly on the x-axis.

=====
SOLUTION
-----
Root (x) = 1.00
x-intercept: (1.00, 0)
=====
Process finished with exit code 0
```

PASSED

Test Case 03: Negative discriminant

Input values used: a = 1, b = 2, c = 5

Expected decision or mathematical result: Discriminant = -16 (NEGATIVE), so two complex/imaginary roots. No solutions or x-intercepts are printed. Vertex is (-1.00, 4.00). The parabola does not touch the x-axis.

Actual program output:

```
Python
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 1
b: 2
c: 5

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
```

```

=====
EQUATION
1.00x^2 + 2.00x + 5.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (-1.00, 4.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: -16.00
Classification: NEGATIVE
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: TWO COMPLEX/IMAGINARY ROOTS
Geometric Interpretation: The parabola does not touch or cross the x-axis.
=====

```

```

Run CC03_Challenge02_QuadraticAnalyzer x
/Users/francisobille/Documents/Coding-Challenge03-APPPAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPPAET-/CC03_Challenge02_QuadraticAr
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 1
b: 2
c: 5

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
1.00x^2 + 2.00x + 5.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (-1.00, 4.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: -16.00
Classification: NEGATIVE
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: TWO COMPLEX/IMAGINARY ROOTS
Geometric Interpretation: The parabola does not touch or cross the x-axis.

```

PASSED

Test Case 04: Upward-opening parabola

Input values used: $a = 2$, $b = -4$, $c = -6$

Expected decision or mathematical result: Direction is UPWARD and vertex type is MINIMUM. Vertex is (1.00, -8.00). Discriminant = 64, with roots $x_1 = 3.00$ and $x_2 = -1.00$.

Actual program output:

Python

Enter coefficients for the quadratic equation ($ax^2 + bx + c = 0$):

a: 2

b: -4

c: -6

=====

QUADRATIC EQUATION AND PARABOLA ANALYZER

=====

EQUATION

$2.00x^2 + -4.00x + -6.00 = 0$

PARABOLA PROPERTIES

Opening Direction: UPWARD

Vertex Type: MINIMUM

Vertex Coordinates: (1.00, -8.00)

DISCRIMINANT ANALYSIS

Discriminant Value: 64.00

Classification: POSITIVE

ROOTS CLASSIFICATION

Number of Solutions: TWO DISTINCT REAL ROOTS

Geometric Interpretation: The parabola crosses the x-axis twice.

SOLUTIONS

Root 1 (x_1) = 3.00

Root 2 (x_2) = -1.00

x-intercepts: (3.00, 0) and (-1.00, 0)

=====

```
Run CC03_Challenge02_QuadraticAnalyzer
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge02_QuadraticA
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: 2
b: -4
c: -6

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
2.00x^2 + -4.00x + -6.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.00, -8.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 64.00
Classification: POSITIVE
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: TWO DISTINCT REAL ROOTS
Geometric Interpretation: The parabola crosses the x-axis twice.

=====
EXPORT.JSON
2.00x^2 + -4.00x + -6.00 = 0
-----
PARABOLA PROPERTIES
-----
Opening Direction: UPWARD
Vertex Type: MINIMUM
Vertex Coordinates: (1.00, -8.00)
-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 64.00
Classification: POSITIVE
-----
ROOTS CLASSIFICATION
-----
Number of Solutions: TWO DISTINCT REAL ROOTS
Geometric Interpretation: The parabola crosses the x-axis twice.
-----
SOLUTIONS
-----
Root 1 (x1) = 3.00
Root 2 (x2) = -1.00
x-intercepts: (3.00, 0) and (-1.00, 0)
=====
```

PASSED

Test Case 05: Downward-opening parabola

Input values used: a = -1, b = 4, c = -3

Expected decision or mathematical result: Direction is DOWNWARD and vertex type is MAXIMUM. Vertex is (2.00, 1.00). Discriminant = 4, with roots x1 = 1.00 and x2 = 3.00.

Actual program output:

```
Python
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: -1
b: 4
c: -3

=====
```

QUADRATIC EQUATION AND PARABOLA ANALYZER

EQUATION

$$-1.00x^2 + 4.00x + -3.00 = 0$$

PARABOLA PROPERTIES

Opening Direction: DOWNWARD

Vertex Type: MAXIMUM

Vertex Coordinates: (2.00, 1.00)

DISCRIMINANT ANALYSIS

Discriminant Value: 4.00

Classification: POSITIVE

ROOTS CLASSIFICATION

Number of Solutions: TWO DISTINCT REAL ROOTS

Geometric Interpretation: The parabola crosses the x-axis twice.

SOLUTIONS

Root 1 (x1) = 1.00

Root 2 (x2) = 3.00

x-intercepts: (1.00, 0) and (3.00, 0)

```
Run CC03_Challenge02_QuadraticAnalyzer
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge02_Quadratic
Enter coefficients for the quadratic equation (ax^2 + bx + c = 0):
a: -1
b: 4
c: -3

=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
EQUATION
-1.00x^2 + 4.00x + -3.00 = 0

PARABOLA PROPERTIES
-----
Opening Direction: DOWNWARD
Vertex Type: MAXIMUM
Vertex Coordinates: (2.00, 1.00)

-----
DISCRIMINANT ANALYSIS
-----
Discriminant Value: 4.00
Classification: POSITIVE

-----
ROOTS CLASSIFICATION
-----
Number of Solutions: TWO DISTINCT REAL ROOTS
Geometric Interpretation: The parabola crosses the x-axis twice.
```

```
Run CC03_Challenge02_QuadraticAnalyzer
-1.00x^2 + 4.00x + -3.00 = 0
=====
PARABOLA PROPERTIES
=====
Opening Direction: DOWNWARD
Vertex Type: MAXIMUM
Vertex Coordinates: (2.00, 1.00)
=====
DISCRIMINANT ANALYSIS
=====
Discriminant Value: 4.00
Classification: POSITIVE
=====
ROOTS CLASSIFICATION
=====
Number of Solutions: TWO DISTINCT REAL ROOTS
Geometric Interpretation: The parabola crosses the x-axis twice.
=====
SOLUTIONS
=====
Root 1 (x1) = 1.00
Root 2 (x2) = 3.00
x-intercepts: (1.00, 0) and (3.00, 0)
=====
```

PASSED

Test Case 06: Invalid quadratic input where a = 0

Input values used: a = 0, b = 5, c = 3

Expected decision or mathematical result: The program prints INVALID INPUT: "The coefficient 'a' cannot be zero. This does not represent a valid quadratic equation." No calculations are shown.

Actual program output:

Python

Enter coefficients for the quadratic equation ($ax^2 + bx + c = 0$):

a: 0

b: 5

c: 3

```
=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
INVALID INPUT
The coefficient 'a' cannot be zero.
This does not represent a valid quadratic equation.
=====
```

```
Run CC03_Challenge02_QuadraticAnalyzer
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge02_QuadraticA
Enter coefficients for the quadratic equation (x^2 + bx + c = 0):
a: 0
b: 5
c: 3
=====
QUADRATIC EQUATION AND PARABOLA ANALYZER
=====
INVALID INPUT
The coefficient 'a' cannot be zero.
This does not represent a valid quadratic equation.
=====
```

PASSED

III. C03: Linear System Analyzer

Test Case 01: One unique solution

Input values used: $a_1 = 2$, $b_1 = 1$, $c_1 = 5$, $a_2 = 1$, $b_2 = -1$, $c_2 = 1$

Expected decision or mathematical result: $D = -3$, $D_x = -6$, $D_y = -3$. CONSISTENT AND INDEPENDENT, with $x = 2.00$ and $y = 1.00$. Verification: $2(2) + 1(1) = 5$ and $1(2) - 1(1) = 1$, so both checks PASS.

Actual program output:

```
Python
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 2
b1: 1
c1: 5

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1
b2: -1
c2: 1

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
2.00x + 1.00y = 5.00

EQUATION 2
1.00x + -1.00y = 1.00
-----
DETERMINANT ANALYSIS
-----
D = -3.00
Dx = -6.00
Dy = -3.00
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.
-----
SOLUTION
-----
x = 2.00
```

y = 1.00

Intersection Point: (2.00, 1.00)

VERIFICATION

Equation 1: 2.00(2.00) + 1.00(1.00) = 5.00 PASSED

Equation 2: 1.00(2.00) + -1.00(1.00) = 1.00 PASSED
=====

```
Run CC03_Challenge03_LinearSystemAnalyzer
/Users/francisobille/Documents/Coding-Challenge03-APPOAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge03_LinearSystemAnalyzer.py
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 2
b1: 1
c1: 5
Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1
b2: -1
c2: 1
=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
2.00x + 1.00y = 5.00
EQUATION 2
1.00x + -1.00y = 1.00
-----
DETERMINANT ANALYSIS
-----
D = -3.00
Dx = -6.00
Dy = -3.00
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.
-----
SOLUTION
-----
x = 2.00
y = 1.00
Intersection Point: (2.00, 1.00)
-----
VERIFICATION
-----
Equation 1: 2.00(2.00) + 1.00(1.00) = 5.00 PASSED
Equation 2: 1.00(2.00) + -1.00(1.00) = 1.00 PASSED
=====
```

PASSED

Test Case 02: No solution

Input values used: a1 = 1, b1 = 1, c1 = 2, a2 = 2, b2 = 2, c2 = 6

Expected decision or mathematical result: D = 0, Dx = -2, Dy = 2. INCONSISTENT, with no solution. The lines are parallel and distinct. No solution or verification section is printed.

Actual program output:

```
Python
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 1
b1: 1
c1: 2

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 2
b2: 2
c2: 6

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====

EQUATION 1
1.00x + 1.00y = 2.00

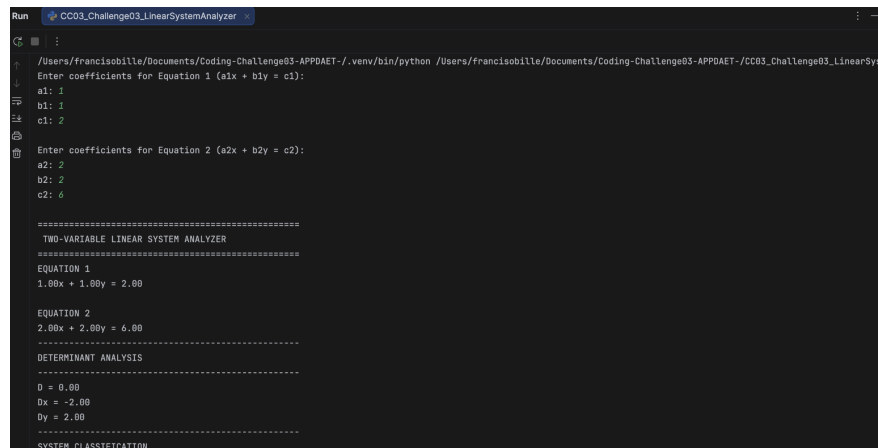
EQUATION 2
2.00x + 2.00y = 6.00
-----

DETERMINANT ANALYSIS
-----

D = 0.00
Dx = -2.00
Dy = 2.00
-----

SYSTEM CLASSIFICATION
-----

Classification: INCONSISTENT
Number of Solutions: NO SOLUTION
Geometric Interpretation: The lines are parallel and distinct.
=====
```



```
Run CCO3_Challenge03_LinearSystemAnalyzer x
/Users/francisobille/Documents/Coding-Challenge03-APPDAET-/venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge03_LinearSys
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 1
b1: 1
c1: 2

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 2
b2: 2
c2: 6

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====

EQUATION 1
1.00x + 1.00y = 2.00

EQUATION 2
2.00x + 2.00y = 6.00
-----

DETERMINANT ANALYSIS
-----

D = 0.00
Dx = -2.00
Dy = 2.00
-----

SYSTEM CLASSIFICATION
-----
```

```
Run CC03_Challenge03_LinearSystemAnalyzer
b2: 2
c2: 6

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
1.00x + 1.00y = 2.00

EQUATION 2
2.00x + 2.00y = 6.00

-----
DETERMINANT ANALYSIS
-----
D = 0.00
Dx = -2.00
Dy = 2.00

-----
SYSTEM CLASSIFICATION
-----
Classification: INCONSISTENT
Number of Solutions: NO SOLUTION
Geometric Interpretation: The lines are parallel and distinct.
=====
```

PASSED

Test Case 03: Infinitely many solutions

Input values used: $a_1 = 1$, $b_1 = 1$, $c_1 = 2$, $a_2 = 2$, $b_2 = 2$, $c_2 = 4$

Expected decision or mathematical result: $D = 0$, $D_x = 0$, $D_y = 0$. CONSISTENT AND DEPENDENT, with infinitely many solutions. The equations describe the same line.

Actual program output:

Python

Enter coefficients for Equation 1 ($a_1x + b_1y = c_1$):

a_1 : 1

b_1 : 1

c_1 : 2

Enter coefficients for Equation 2 ($a_2x + b_2y = c_2$):

a_2 : 2

b_2 : 2

c_2 : 4

=====

TWO-VARIABLE LINEAR SYSTEM ANALYZER

=====

EQUATION 1

$1.00x + 1.00y = 2.00$

EQUATION 2

$2.00x + 2.00y = 4.00$

DETERMINANT ANALYSIS

 $D = 0.00$

$D_x = 0.00$

$D_y = 0.00$

SYSTEM CLASSIFICATION

Classification: CONSISTENT AND DEPENDENT

Number of Solutions: INFINITELY MANY SOLUTIONS

Geometric Interpretation: The equations describe the same line.

=====

```
Run CC03_Challenge03_LinearSystemAnalyzer
/Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge03_LinearSys
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 1
b1: 1
c1: 2

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 2
b2: 2
c2: 4

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
1.00x + 1.00y = 2.00

EQUATION 2
2.00x + 2.00y = 4.00
-----
DETERMINANT ANALYSIS
-----
D = 0.00
Dx = 0.00
Dy = 0.00
-----
SYSTEM CLASSIFICATION
```

```
Run CC03_Challenge03_LinearSystemAnalyzer
a1: 1
b1: 1
c1: 2

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
1.00x + 1.00y = 2.00

EQUATION 2
2.00x + 2.00y = 4.00
-----
DETERMINANT ANALYSIS
-----
D = 0.00
Dx = 0.00
Dy = 0.00
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND DEPENDENT
Number of Solutions: INFINITELY MANY SOLUTIONS
Geometric Interpretation: The equations describe the same line.
=====
```

PASSED

Test Case 04: Negative coefficients

Input values used: $a_1 = -1$, $b_1 = 2$, $c_1 = -5$, $a_2 = 3$, $b_2 = -1$, $c_2 = 5$

Expected decision or mathematical result: $D = -5$, $D_x = -5$, $D_y = 10$. CONSISTENT AND INDEPENDENT, with $x = 1.00$ and $y = -2.00$. Verification: $-1(1) + 2(-2) = -5$ and $3(1) - 1(-2) = 5$, so both checks PASS.

Actual program output:

```
Python
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: -1
b1: 2
c1: -5

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 3
b2: -1
c2: 5

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
-1.00x + 2.00y = -5.00

EQUATION 2
3.00x + -1.00y = 5.00
-----
DETERMINANT ANALYSIS
-----
D = -5.00
Dx = -5.00
Dy = 10.00
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.
-----
SOLUTION
-----
x = 1.00
y = -2.00
Intersection Point: (1.00, -2.00)
-----
```

VERIFICATION

Equation 1: $-1.00(1.00) + 2.00(-2.00) = -5.00$ PASSED

Equation 2: $3.00(1.00) + -1.00(-2.00) = 5.00$ PASSED

=====

```
Run CC03_Challenge03_LinearSystemAnalyzer
/Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobilla/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge03_LinearSystemAnalyzer.py
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: -1
b1: 2
c1: -5
Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 3
b2: -1
c2: 5
=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
-1.00x + 2.00y = -5.00
|
EQUATION 2
3.00x + -1.00y = 5.00
-----
DETERMINANT ANALYSIS
-----
D = -5.00
Dx = -5.00
Dy = 10.00
-----
```

```
Run CC03_Challenge03_LinearSystemAnalyzer
0.00x + -1.00y = 0.00
-----
DETERMINANT ANALYSIS
-----
D = -5.00
Dx = -5.00
Dy = 10.00
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.
-----
SOLUTION
-----
x = 1.00
y = -2.00
Intersection Point: (1.00, -2.00)
-----
VERIFICATION
-----
Equation 1: -1.00(1.00) + 2.00(-2.00) = -5.00 PASSED
Equation 2: 3.00(1.00) + -1.00(-2.00) = 5.00 PASSED
=====
```

PASSED

Test Case 05: Decimal coefficients

Input values used: a1 = 0.5, b1 = 0.25, c1 = 1.75, a2 = 1.5, b2 = -0.5, c2 = 1.5

Expected decision or mathematical result: Exact values are D = -0.625, Dx = -1.25, Dy = -1.875. CONSISTENT AND INDEPENDENT, with x = 2.00 and y = 3.00. Verification gives 1.75 and 1.50, so both checks PASS. Python's `:.2f` rounding will display D as -0.62 and Dy as -1.88, so don't be surprised by that.

Actual program output:

```
Python
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 0.5
b1: 0.25
c1: 1.75

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1.5
b2: -0.5
c2: 1.5

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
0.50x + 0.25y = 1.75

EQUATION 2
1.50x + -0.50y = 1.50
-----
DETERMINANT ANALYSIS
-----
D = -0.62
Dx = -1.25
Dy = -1.88
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.
-----
SOLUTION
-----
x = 2.00
y = 3.00
Intersection Point: (2.00, 3.00)
-----
VERIFICATION
-----
Equation 1: 0.50(2.00) + 0.25(3.00) = 1.75 PASSED
Equation 2: 1.50(2.00) + -0.50(3.00) = 1.50 PASSED
=====
```

```
Run CC03_Challenge03_LinearSystemAnalyzer
/Users/francisobilla/Documents/Coding-Challenge03-APPOAET-/venv/bin/python /Users/francisobilla/Documents/Coding-Challenge03-APPOAET-/CC03_Challenge03_LinearSystemAnalyzer.py
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 0.5
b1: 0.25
c1: 1.75

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1.5
b2: -0.5
c2: 1.5

=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
EQUATION 1
0.50x + 0.25y = 1.75

EQUATION 2
1.50x + -0.50y = 1.50

-----
DETERMINANT ANALYSIS
-----
D = -0.62
Dx = -1.25
Dy = -1.88
-----
SYSTEM CLASSIFICATION
-----
Classification: CONSISTENT AND INDEPENDENT
Number of Solutions: ONE UNIQUE SOLUTION
Geometric Interpretation: The lines intersect.

-----
SOLUTION
-----
x = 2.00
y = 3.00
Intersection Point: (2.00, 3.00)

-----
VERIFICATION
-----
Equation 1: 0.50(2.00) + 0.25(3.00) = 1.75 PASSED
Equation 2: 1.50(2.00) + -0.50(3.00) = 1.50 PASSED
=====
```

PASSED

Test Case 06: Invalid equation input

Input values used: a1 = 0, b1 = 0, c1 = 5, a2 = 1, b2 = 1, c2 = 2

Expected decision or mathematical result: The program prints INVALID INPUT: "Equation 1 is invalid: both a1 and b1 are zero." Equation 2 is valid, so isn't flagged. No determinants or solutions are computed.

Actual program output:

```
Python
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 0
b1: 0
c1: 5

Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1
b2: 1
c2: 2

=====
```

TWO-VARIABLE LINEAR SYSTEM ANALYZER

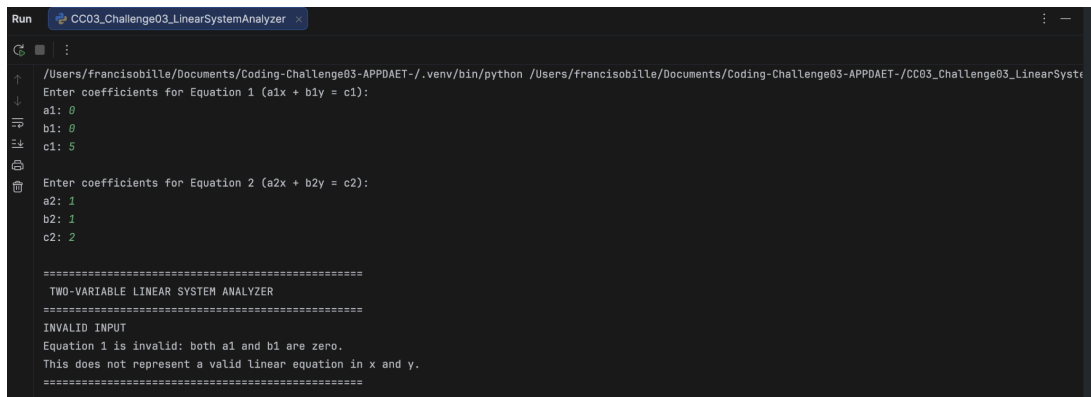
=====

INVALID INPUT

Equation 1 is invalid: both a1 and b1 are zero.

This does not represent a valid linear equation in x and y.

=====



```
Run CC03_Challenge03_LinearSystemAnalyzer x
/Users/francisobille/Documents/Coding-Challenge03-APPDAET-/.venv/bin/python /Users/francisobille/Documents/Coding-Challenge03-APPDAET-/CC03_Challenge03_LinearSystemAnalyzer.py
Enter coefficients for Equation 1 (a1x + b1y = c1):
a1: 0
b1: 0
c1: 5
Enter coefficients for Equation 2 (a2x + b2y = c2):
a2: 1
b2: 1
c2: 2
=====
TWO-VARIABLE LINEAR SYSTEM ANALYZER
=====
INVALID INPUT
Equation 1 is invalid: both a1 and b1 are zero.
This does not represent a valid linear equation in x and y.
=====
```

PASSED